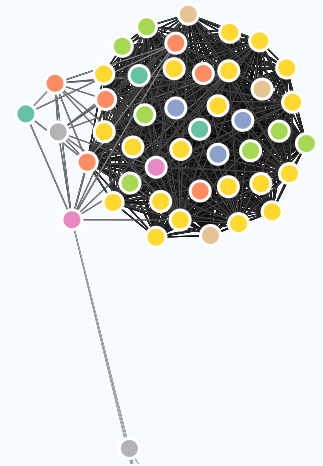


JONATHAN BERRISCH

I specialized in econometrics for energy forecasting and am well-versed in R, C++, and Python. I currently work in academia between statistics and information technology. I research probabilistic forecasting and maintain a growing IT infrastructure with databases. I also actively contribute to open-source projects and disseminate knowledge by creating beautiful visualizations.



View this CV online with links at berrisch.biz/cv

EDUCATION

- 2025
|
2020
- **PhD, Econometrics**
University of Duisburg Essen 📍 Essen
 - Data Science Methods for Forecasting in Energy and Economics
 - FOSS Projects: [profoc](#), [ondil](#), [RcppTimer](#), [CppTimer](#), [dccpp](#)
 - Graduated with Summa Cum Laude
- 2020
|
2017
- **M.Sc., Economics**
University of Duisburg Essen 📍 Essen
 - Focused on Econometrics
 - Gathered experience in R, Python and Git
 - Graduated with Distinction: 1.2
- 2017
|
2013
- **B.Sc. Economics and Business Administration**
FernUniversität in Hagen 📍 Remote
 - Focused on Monetary Policy and Quantitative Economics
 - Final Grade: 1.7

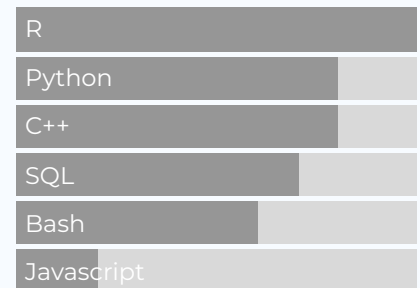
PERSONAL

🏠 1992
📍 Wuppertal, Germany
✉ Jonathan@Berrisch.biz
☎ +49 172 2173741
🌐 berrisch.biz
🎧 [BerriJ](#)
📖 [BerriJ](#)
🎓 [University](#)

SOFTWARE PROJECTS

- 2024
- **[ondil](#)**
A Python Package for Online Distributional Regression
 - Provides online estimation of models for distributional regression, respectively, models for conditional heteroskedastic data.
 - Uses [numba](#) just in time compilation to improve performance.
- 2024
- **[rcpptimer](#)**
An R package for Benchmarking C++ Code
 - Provides bindings for [cpptimer](#), a tic-toc class for timing C++ code
 - Supports nested and overlapping timers and OpenMP parallelism
 - Bridges the gap between R and C++ by offering a simple and efficient solution for timing code within the Rcpp ecosystem

LANGUAGE SKILLS



DEVELOPMENT TOOLS



2021



profoc

An R Package for Probabilistic Forecast Combination

- An R package to combine probabilistic forecasts using the CRPS learning algorithm introduced in [Berrisch, Ziel, 2023](#)
- The main algorithms are implemented in C++
- Bindings to R are realized using Rcpp



RESEARCH EXPERIENCE

Current
|
2025



Postdoctoral Researcher

University of Duisburg Essen

📍 Essen

- Probabilistic Forecasting in Energy Markets
- FOSS Projects: [entsoe-apy](#) [nixos-config](#)
- Managing NixOS deployments

2025
|
2020



Doctoral Researcher

University of Duisburg Essen

📍 Essen

- Chair of Environmental Economics and Renewable Energies Prof. Dr. Ziel
- Development and maintenance of Energymarket databases (MySQL, R, Python, Docker), which are used within and outside our department.
- Maintenance of the IT infrastructure (Arch, NixOS, Ubuntu)

2019
|
2018



Research Aide

University of Duisburg Essen

📍 Essen

- Chair of Econometrics: Prof. Dr. Christoph Hanck
- Working with R, Latex, Git, SQL
- First teaching experience



INDUSTRY EXPERIENCE

2020
|
2020



Masters Degree Candidate

RWE Supply and Trading GmbH

📍 Essen

- Developing forecasting models for European natural gas prices
- Working with Python, Tensorflow, SQL, Git
- Published [studentt](#)

2017
|
2015



Working Student

[Stepstone.de](#)

📍 Duesseldorf

- Quality Assurance

2015
|
2014



Customer Service Representative

Stadtsparkasse Duesseldorf

📍 Duesseldorf

2014
|
2012



Banking Apprentice

Stadtsparkasse Duesseldorf

📍 Duesseldorf

I gathered industry experience early by studying the bachelors to moonlight.



PUBLICATIONS

- 2025 ● **rcpptimer: Rcpp Tic-Toc Timer with OpenMP Support**
J. Berrisch - [arXiv Preprint](#)
- 2024 ● **Online Distributional Regression**
S. Hirsch, J. Berrisch, F. Ziel - [arXiv Preprint](#)
- 2024 ● **Multivariate Probabilistic CRPS Learning with an Application to Day-Ahead Electricity Prices**
J. Berrisch, F. Ziel - [International Journal of Forecasting](#)
- 2023 ● **CRPS Learning**
J. Berrisch, F. Ziel - [Journal of Econometrics](#)
- 2023 ● **Modeling volatility and dependence of European carbon and energy prices**
J. Berrisch, S. Pappert, A. Arsova, F. Ziel - [Finance Research Letters](#)
- 2023 ● **High-resolution peak demand estimation using generalized additive models and deep neural networks**
J. Berrisch, F. Ziel - [Energy and AI](#)
- 2022 ● **Distributional modeling and forecasting of natural gas prices**
J. Berrisch, F. Ziel - [Journal of Forecasting](#)

CONFERENCES AND WORKSHOPS

- 2025 ● **International Workshop on Statistical Modelling (IWSM)**
Poster Presentation  Limerick, Ireland
· Online Distributional Regression in Python
- 2025 ● **Workshop KI für die Wasserforschung**
Participant  Koblenz
- Brussels, Belgium ● **E-TREND Joint Workshop on Postprocessing, Forecasting and Nowcasting**
for Renewable Energy (RenEUCast)  Speaker
2025
· ENTSO-E API Python Package: Streamlining Access to European Electricity Market Data
- 2024 ● **useR! — International R User Conference**
Speaker  Salzburg, Austria
· Benchmarking (R)Cpp Code with rcpptimer

I like exchanging ideas,
connecting with people and
shaping the future.

2024	● International Symposium on Forecasting Speaker · Evaluating C++ Computation Times with 'rcpptimer'	📍 Dijon, France
2024	● Strommarkttreffen Speaker	📍 Berlin
2024	● Workshop on Market Power in Energy Markets Participant	📍 Berlin
2023	● International Ruhr Energy Conference (INREC) Speaker · High-Resolution Peak Demand Estimation Using Generalized Additive Models and Deep Neural Networks	📍 Essen
2023	● Statistische Woche 2023 Speaker · Modeling Volatility and Dependence of European Carbon and Energy Prices	📍 Dortmund
2023	● ARGE Masterclass 2023 “Econometrics of Energy Markets“ Speaker · Modeling Volatility and Dependence of European Carbon and Energy Prices	📍 Berlin
2023	● ECMI 2023 Conference Speaker · Multivariate Probabilistic CRPS Learning: An Application to Day-Ahead Power Prices	📍 Wrocław, Poland
2023	● Smart4RES Final Conference Participant · Next-generation Forecasting of Renewable Generation for Large-scale Integration in Energy Systems and Markets	📍 Paris, France
2022	● International Symposium on Forecasting Speaker · High-Resolution Peak Demand Estimation Using Generalized Additive Models and Deep Neural Networks · Best Student Presentation Award	📍 Oxford
2022	● UseR! 2022 Conference Speaker · An Introduction to Rcpp Modules	📍 Remote

2022	● International Conference on Computational Finance 2022 Speaker	📍 Wuppertal
	• Multivariate CRPS Learning with Applications to Electricity Price Forecasting	
2022	● 15TH RGS Doctoral Conference in Economics Speaker	📍 Remote
	• CRPS Learning	
2021	● International Symposium on Forecasting Speaker	📍 Remote
	• CRPS Learning	
2021	● European conference on Operational Research Speaker	📍 Remote
	• CRPS Learning	
2020	● International Symposium on Forecasting Speaker	📍 Remote
	• Modeling and Probabilistic Forecasting of Natural Gas Prices	
2020	● International Ruhr Energy Conference (INREC) Speaker	📍 Remote
	• Data Science Methods for Probabilistic Natural Gas Price Forecasting	



VOLUNTARY

Current 2024	● Member of the Faculty Council University of Duisburg Essen	📍 Essen
	• Faculty of Business Administration and Economics	
Current 2023	● Member of the Executive Board /dev/tal e.V. Hackerspace	📍 Wuppertal
2022 2020	● Member of the Student Parliament University of Duisburg Essen	📍 Essen
	• Member of the Presidium (2020 - 2021)	
2020 2019	● Committees University of Duisburg Essen	📍 Essen
	• Evaluation committee for Jun. Prof. Dr. Ziel (Student Member) • Evaluation committee for Jun. Prof. Dr. Beck (Student Member) • Appointments committee for a new professorship in economics esp. public finance (Student Member)	

I'm excited about connecting the dots and putting technology to good use.

2018



Mentor

University of Duisburg Essen

- Supporting the elementary students of M.Sc. Economics

📍 Essen

2018



Mentor

University of Duisburg Essen

- Supporting the elementary students of M.Sc. Economics

📍 Essen



INTERESTS



Cooking

Pad-Thai, Sourdough Bread, Lasagne, Cookies, etc.



Cycling

In everyday life and on vacation



Information Technology

At Home, at the Hackspace, etc.

- Self-Hosting: [Gitea](#), [Tandoor](#), [Home Assistant](#), [Wanderer](#)
- Building a [Voron 2.4](#)
- Fiddling around with hardware and software

Working in IT and spending time in a hackerspace is a perfect match.

Last updated on 2025-12-03.

Source: github.com/BerriJ/cv.